S4 Second Internal Examination, July 2023

19-205-0404:APPLIED THERMODYNAMICS

Time: 2 Hours Maximum Marks :40

Course Outcomes:

1. Learn first and second law of thermodynamics and the application of these laws for various thermodynamic processes.
2. Apply the knowledge of thermodynamic laws to various engineering systems.
3. Analyse the performance of steam generators with additional accessories used for enhancing the performance.
4. Understand the various types of nozzles and its performance at various back pressure conditions and also for different flow conditions.
5. Calculate the mole and mass fractions of various mixtures and also various specific properties of the mixtures.

Use separate answer books for Module 1&2 and 3&4

Module – 3 & 4		**Marks**	**BL**	**CO**
1	Two vessels A and B, both containing Nitrogen is connected by a valve which is opened and allowed the contents to mix. Vessel A contains 16.8 kg of N_2 at 16 bar pressure & 55 ^{0}C temp. and Vessel B contains 3 kg of N_2 at 6.4 bar pressure & 25 ^{0}C temp. Find the final temperature and pressure.	6	4	5
2	Explain the super saturated flow in steam nozzles	4	2	4
3	What are the effects of Friction in a steam nozzle?	4	1	4
4	In a convergent divergent nozzle, steam at 11 bar and 200 ^{0}C temp is expanded to 0.1 bar. If the outlet area of the nozzle is 10 cm^2, determine the mass of steam discharged per hour.	6	4	4
Module – 1& 2				
5	A Carnot cycle works with adiabatic compression ratio of 5 and isothermal expansion ratio of 2. The volume of air at the beginning of isothermal expansion is 0.3 m^3. If the maximum temperature and pressure is limited to 550 K and 21 bar, determine (i) Maximum temperature in the cycle (ii) Thermal efficiency of the cycle (iii) Pressure at all salient points (iv) Work done per cycle Take $\gamma = 1.4$	8	4	1
6	Write short note on thermodynamic temperature scales. What is the need for different scales of temperature?	5	2	1
7	Saturated water at 10 bar is fed to a heat exchanger at 5 kg/min. It is heated isobarically to 300 ^{0}C. a. Calculate the final state of water b. Calculate the change in enthalpy in kW c. Compare the volumetric flow rates.	7	4	2